

BI2025 Experiment Report – Group 059

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Abstract

This report documents a machine learning experiment conducted by Group 059 following the CRISP-DM process model. The study focuses on identifying key institutional factors that influence student graduation rates and developing predictive models to support data-driven decision making in higher education.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

CRISP-DM, Provenance, Knowledge Graph, Machine Learning, Graduation Rate Prediction, Institutional Performance

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1 Business Understanding

1.1 Data Source and Scenario

The colleges_usnews dataset used for this project is taken from OpenML. It is a real-world dataset, from the American Statistical Association’s 1995 Data Analysis Exposition, based on institutional data published in U.S News and World Report’s Guide to America’s Best Colleges (1993-1994). It contains information for 1,302 U.S colleges and universities, including variables related to admission selectivity (SAT/ACT scores, application acceptance), institutional characteristics (public/private, tuition, spending), student composition, faculty qualification and outcomes such as graduation rate. In total the dataset has a total of 35 attributes with most of them being numeric, and some being categorical such as state, public/private. The dataset contains missing values, particularly in the test score variables making it a realistic analysis.

A suitable business scenario for this dataset would be a university administration seeking to understand what factors influence graduation rates. By applying data analytics processes to the dataset, the

institution can identify patterns and understand what factors influence a higher graduation rate. This analysis can support strategic planning, resource allocation and performance benchmarking.

1.2 Business Objectives

The main objective of this analysis is to define which of the variables have a higher impact on the graduation rate of students. By identifying these factors the institutions have the necessary information to improve students success. Other objectives include providing universities with benchmarks against similar institution, identifying what aspects would have the biggest turn on investments and understanding how admission selectivity or student preparedness relate to graduation outcomes.

All these objectives support organizations in making informed strategic decisions.

1.3 Success Criteria

For this project to be called successful, it should produce precise and easily interpretable insights regarding the key factors that influence the graduation rate as well as a model that accurately predicts the graduation rate of an institution based on the provided features. Another measure of success for this analysis would be that the derived recommendations are actionable, such as identifying which variables universities should influence to improve outcomes.

Additionally, the results should be presented in a way that universities can easily use the results for benchmarking, this way they can easily compare their performance against other educational institutions.

1.4 Data Mining Goals

The main data mining goal is to build a predictive model that estimates an institution’s graduation rate based on the available institutional, financial and student-related variables in the dataset. The model should provide accurate predictions, as well as highlight which features contribute most strongly to higher or lower graduation rates. Another goal is to perform exploratory analysis to better understand the relationships between variables such as tuition, selectivity, spending per student, or faculty resources and the graduation rate. This supports the business objective of identifying concrete levers that universities can influence to improve student success.

1.5 Data Mining Success Criteria

From a data mining perspective, the project is considered successful if we can train a regression model that generalizes well and explains an important part of the variance in graduation rates. We are trying to have a model that achieves a high coefficient of determination and a low prediction error when testing via cross-validation. In addition, the learned model and derived feature importance measures should be interpretable for non-technical stakeholders. Key factors

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influencing graduation rate should be distinctly identifiable (similar across different validation splits), so that they can be translated into recommendations and benchmarking indicators for universities.

1.6 AI Risk Aspects

This project uses institutional-level, historical data about colleges and universities. While the dataset does not contain personal student identifiers, there are still some AI-related risks to consider. First, the data may reflect existing structural inequalities (between public and private institutions or between different regions). A model trained on such data could unintentionally reinforce these patterns if its predictions are used for funding decisions, rankings or policy making without careful interpretation. Second, there is a risk of over-interpreting correlations as causal effects and using the model as a black-box decision tool rather than as analytic support. To handle these risks, the focus of the project is on transparency and being able to interpret the data (using explainable models and feature importance analysis), precisely communicating the limitations of the data and the model, and treating the results as decision-support for experts rather than automatic decision-making.

2 Data Understanding

2.1 Dataset Description

The dataset contains U.S. colleges and universities derived from US News and World Report. Each row represents one institution and the columns describe selectivity, resources, financial indicators, and student outcomes such as graduation rate.

2.2 Raw Data Features

Table 1 lists the main attributes of the dataset together with their types and sematic meaning. For clarity and readability, only the most relevant features for the analysis are listed here the complete schema of the dataset is documented in the provenance graph.

Table 1: Raw Data Features

Feature	Type	Description
Additional_fees	numeric	Mandatory non-tuition fees.
Average_ACT_score	numeric	Mean ACT score of admitted students.
Graduation_rate	numeric	Percentage of students graduating.
Public/private_indicator	categorical	Institution control type.
State	categorical	Institution location (U.S. state).
Student/faculty_ratio	numeric	Students per faculty member.
Pct.alumni_who_donate	numeric	Alumni donation rate.
Instructional_expenditure_per_student	numeric	Instructional spending per student.
Third_quartile-Math_SAT	numeric	75th percentile Math SAT score.
Average_Math_SAT_score	numeric	Mean Math SAT score.
Average_Verbal_SAT_score	numeric	Mean Verbal SAT score.
Average_Combined_SAT_score	numeric	Mean combined SAT score.
First_quartile-Math_SAT	numeric	25th percentile Math SAT score.
First_quartile-Verbal_SAT	numeric	25th percentile Verbal SAT score.
Third_quartile-Verbal_SAT	numeric	75th percentile Verbal SAT score.
First_quartile-ACT	numeric	25th percentile ACT score.
Third_quartile-ACT	numeric	75th percentile ACT score.
Number_of_applications_received	numeric	Total applications received.
Number_of_applicants_accepted	numeric	Total applicants accepted.
Number_of_new_students_enrolled	numeric	Newly enrolled students.

2.3 Descriptive Statistics

The descriptive statistics for all attributes were computed during the Data Understanding phase. The Table 2 shows a summary of all the data characteristics for the key variables.

Table 2: Descriptive Statistics of Key Variables

Variable	Count	Mean	Std.	Min-Max
Public/private indicator	1302	1.64	0.48	1–2
Average Math SAT score	777	506.8	67.8	320–750
Average Verbal SAT score	777	461.2	58.3	280–665
Average Combined SAT score	779	968.0	123.6	600–1410
Average ACT score	714	22.1	2.6	11–31
Applications received	1292	2752	3542	35–48094
Applicants accepted	1291	1871	2251	35–26330
New students enrolled	1297	779	885	18–7425
Pct. new students top 10% HS	1067	25.7	18.3	1–98
In-state tuition	1272	7897	5348	480–25750

2.4 Data Quality Assessment

A systematic evaluation of data quality was conducted. Missing values were assessed across all attributes, revealing that 31 variables contain at least one missing observation (see Table 3). In addition, potential outliers were investigated using the interquartile range (IQR) rule, identifying values lying outside the interval $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$, based on the descriptive statistics reported in Table 2 and in Table 4 we can see the list of outliers.

Table 3: Number of Missing Values per Attribute

Attribute	Missing Values
Third quartile ACT	639
First quartile ACT	639
Average ACT score	588
First quartile Math SAT	530
First quartile Verbal SAT	530
Third quartile Math SAT	530
Third quartile Verbal SAT	530
Average Math SAT score	525
Average Verbal SAT score	525
Average Combined SAT score	523
Board costs	498
Room costs	321
Additional fees	274
Pct. new students top 10% HS	235
Pct. alumni who donate	222
Pct. new students top 25% HS	202
Estimated personal spending	181
Graduation rate	98
Room and board costs	76
Estimated book costs	48
Instructional expenditure per student	39
Part-time undergraduates	32
Pct. of faculty with Ph.D.s	32
Pct. of faculty with terminal degree	30
In-state tuition	30
Out-of-state tuition	20
Applicants accepted	11
Applications received	10
New students enrolled	5
Full-time undergraduates	3

Table 4: Outliers for Selected Attributes

Attribute	Missing Values
Full-time undergraduates	124
Applications received	121
Part-time undergraduates	119
Applicants accepted	105
New students enrolled	103
Estimated book costs	95
Instructional expenditure per student	84
Pct. new students top 10% HS	60
Additional fees	55
Room costs	47
First quartile Verbal SAT	35
Average Verbal SAT score	28
Estimated personal spending	24
Student/faculty ratio	22
Average Combined SAT score	20

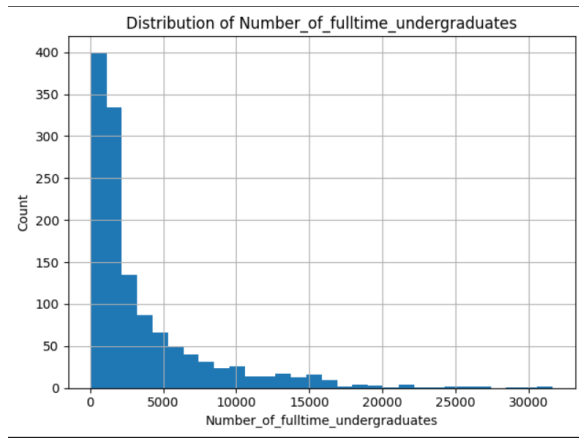


Figure 1: Distribution of full-time undergraduates

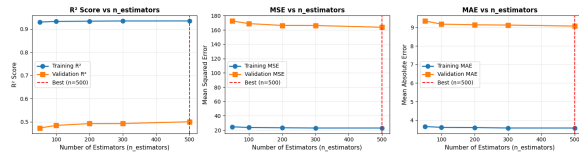


Figure 2: Full-time undergraduates VS public/private indicator

2.5 Visual Exploration

Simple visual analytics were generated to understand the distribution of the target variable and its relationships with selected numeric features. Summary from the visual exploration activity: The distribution in Fig 1 is highly right-skewed; this is because, Fig 1 shows that, most institutions have a small number of students, with most of them having fewer than 5,000 students. Other institutions exceed this number, with some of them having around 20,000–30,000 students enrolled. From Fig 2 we see that public institutions enroll more students, while private institutions tend to be much smaller. These public institutions appear as outliers but removing them would discard valuable information. We take into account that this may introduce heteroscedasticity and affect linear assumptions, but the model remains suitable for exploratory analysis.

2.6 Ethical and Sensitive Aspects (2e)

Potential sensitive attributes include variables related to location, selectivity, or resource levels that might indirectly give us socio-economic status. Analyses and models based on this data risk reproducing existing inequalities between different types of colleges if these attributes are interpreted uncritically as indicators of quality.

Risks include: (1) reinforcing biases embedded in the original ranking methodology (favoring high-resource institutions); (2) misleading stakeholders if predicted graduation rates are treated as objective truth without considering context and (3) drawing causal conclusions from observing aggregate data. Questions for domain

Table 5: Dataset Preview (Academic Profile and Selectivity – First 5 Institutions)

St.	Type	M-SAT	V-SAT	C-SAT	ACT	Q1-M	Q3-M	Q1-V	Grad.
AK	Pub.	490	482	972	20	440	530	430	15
AK	Priv.	505	460	964	22	460	580	410	39
AL	Priv.	505	460	964	17	460	580	410	40
AL	Pub.	505	460	964	20	460	580	410	55
AL	Priv.	505	460	964	21	460	580	410	51

Table 6: Dataset Preview (Financial, Faculty, and Enrollment Attributes – First 5 Institutions)

Board	Add.	Books	Pers.	%PhD	%Term	Stud./Fac.	Alum.	Instr.
2500	130	800	1500	76	72	119	2	10922
2250	34	500	1162	39	51	95	19	9584
1442	155	500	850	53	53	143	19	7043
1700	300	350	1248	52	56	328	19	3971
2000	124	300	600	72	72	189	8	5883

experts would be: Which variables are acceptable to use for decision-making? Are there known reporting biases or problematic indicators in the underlying US News data?

Likely required data preparation steps: handling missing values (distinguishing between truly missing vs. 'not applicable'), implementing categorical variables appropriately, considering transformations for skewed financial variables (log-transforming expenditures or tuition) and analysing the impact of extreme outliers (very high-cost or very low-graduation-rate institutions) on model stability.

3 Data Preparation

In the data preparation phase we applied several preprocessing steps to ensure that the data is suitable for modeling. First, all the missing values were removed from the `graduate_rate` variable, as these records can not be used in supervised learning. In the other variables, the missing attributes were replaced using the median of each respective variable, a process that limits the skewed distributions and extreme values. Even though media imputation was applied to obtain an initial clean dataset for exploration this step will be repeated inside the modeling pipeline only for the training data to avoid data leakage. After median imputation, public/private institutional indicator was converted from numeric codes to meaningful category labels. Lastly, categorical attributes were converted to a category data type so that they can be correctly handled in the future modeling steps. These preprocessing actions resulted in a cleaned dataset with complete numeric information and clearly defined categorical variables, while preserving the original semantic structure of the data. Table 5 and Table 6 show an overview of the cleaned data.

3.0.1 Other preprocessing steps considered. During the data preparation phase, several additional preprocessing actions were evaluated but ultimately not applied.

- **Outlier removal.** An outlier analysis based on the IQR rule revealed a substantial number of extreme values in several numeric attributes, particularly `Number_of_fulltime_undergraduates` (124 outliers), `Number_of_applications_received` (121 outliers), and `Number_of_applicants_accepted` (105 outliers), as well as

in cost-related variables. Visual inspection using scatter plots by institution type indicated that these values correspond to large public universities or highly selective private institutions. Removing these observations would therefore eliminate meaningful structural variation, and outliers were retained.

- **Feature scaling.** Feature scaling was considered but postponed to the modeling phase. Only certain regression algorithms (e.g., linear regression, support vector regression, and neural networks) require standardized inputs, while tree-based models are scale invariant. Scaling of SAT scores and cost-related variables will therefore be applied conditionally, depending on the selected algorithm.
- **Encoding of high-cardinality categorical variables.** Encoding attributes with many distinct values, such as State, would significantly increase the dimensionality of the dataset. This step was deferred to the modeling phase and will be applied only if the chosen algorithm requires numeric input.
- **Attribute removal.** The removal of attributes with many missing values or limited variance such as the SAT and ACT scores was considered but rejected, as all variables carry relevant semantic information about institutional characteristics.

3.0.2 Possible derived attributes. Based on the exploratory analysis, several derived attributes were identified as potentially useful for modeling.

- A *selectivity ratio*, defined as `Number_of_applications_received` divided by `Number_of_applicants_accepted`, to capture institutional selectivity.
- A *total cost* variable, defined as the sum of `In_state_tuition`, `Room_and_board_costs`, `Additional_fees`, `Estimated_book_costs`, and `Estimated_personal_spending`, representing the overall annual cost of attendance.

These derived features may be incorporated during the modeling phase if they prove useful for predicting graduation rates.

3.0.3 External data sources. Graduation outcomes may also be influenced by factors beyond an institution's internal characteristics. Potential external data sources include state-level socio-economic indicators such as median household income, unemployment rates, and cost-of-living indices. Incorporating such data would allow tuition and expense variables to be normalized across regions. Additionally, high-school graduation and college enrollment rates at the state level could provide context on student preparedness, which may further affect graduation outcomes.

4 Data Modeling

4.1 Identifying suitable data mining algorithms

To predict the graduation rates for this dataset, we considered several algorithms:

1. Linear Regression: Simple, interpretable, but may not capture non-linear relationships
2. Random Forest Regression: Handles non-linear relationships, provides feature importance, robust to outliers
3. Gradient Boosting (XGBoost/GBM): Often achieves high performance but more complex and computationally expensive
4. Support Vector Regression: Can handle non-linear relationships

but less interpretable

After careful consideration we chose Random Forest Regression as the primary algorithm because first, it handles mixed data types well without extensive preprocessing. Second it provides feature importance for interpretability. Third is robust to outliers and missing values. Fourth it provides a good balance between performance and interpretability. Fifth it can capture non-linear relationships between features and graduation rate. Finally is less prone to overfitting than single decision trees.

4.2 Identifying hyperparameters for tuning

The Random Forest algorithm provides several hyperparameters that can be tuned to provide the desired results. One of the key parameters is `n_estimators` which shows the number of trees in the forest; the more trees, the better the results but the slower the processing. Another important parameter is `max_depth` which shows the maximum depth of trees. Additionally, `min_sample_split` shows the minimum samples required to split a node, while `min_sample_leaf` shows the minimum samples required at a leaf node. Finally, `max_features` parameter shows the number of features to consider in order to get the best split.

The parameter that we decided to choose is `n_estimator` because it has a significant impact on model performance, it's computationally efficient to tune because it can test multiple values, and it's less likely to cause overfitting especially compared to `max_depth`. Moreover it provides a good balance between model complexity and training time

To decide the optimal number of trees, the following values were tested: [50, 100, 200, 300, 500].

4.3 Defining Train/Validation/Test splits

To split the data into train/validation/test sets, we first defined a random number split with a `random_state` of 42 for reproducibility. Then we defined that the test set will be 15%, and it was held out for final evaluation only. Additionally, the validation set was also defined to be 15% to be used for hyperparameter tuning. Leaving the training set with 70%.

This 70/15/15 split was chosen to: 1. Provide more training data (70%) to help reduce overfitting 2. Ensure adequate validation data (15%) for reliable hyperparameter selection 3. Maintain an independent test set (15%) for unbiased final performance assessment 4. Ensure reproducibility through fixed `random_state`

There was no need for stratification, as this regression problem has a continuous target; also, because there were no time dependencies identified in the data, the random split is appropriate.

4.4 Model Training

The Random Forest models were trained with `n_estimators` values: [50, 100, 200, 300, 500]. Each model was trained on the training set and evaluated on both training and validation sets. Performance metrics (R^2 , MSE, MAE) were computed for each configuration the results are documented at Table 7. The tuning process shows that performance generally improves with more trees, but with diminishing returns.

Table 7: Random Forest Hyperparameter Tuning Results

n_estimators	Train R ²	Val. R ²	Val. MSE	Val. MAE
50	0.9303	0.4729	172.54	9.35
100	0.9331	0.4843	168.82	9.17
200	0.9341	0.4920	166.30	9.14
300	0.9351	0.4924	166.15	9.12
500	0.9351	0.4998	163.73	9.07

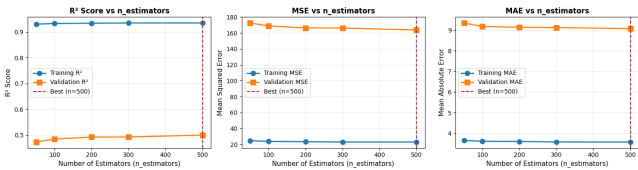


Figure 3: Random Forest Hyperparameter Tuning Results

4.5 Performance Metrics

The most suitable model according to our results is the one with `n_estimators` configuration of 500, as we can see from Fig 3 This model explains approximately 50% of the variance in the validation dataset. In addition it achieves a validation MSE of 163.73 and a validation MAE of 9.07, indicating improved performance compared to other models.

4.6 Retraining the model

During the validation phase, the ideal hyperparameter configuration was identified, and the final Random Forest model was re-trained with the combined training and validation datasets. This step conforms to standard machine learning practice and seeks to maximise the quantity of data available for learning before assessing the model on the independent test set.

Specifically, the training and validation sets were combined, yielding a total of 1,023 samples for final model training. To guarantee methodological consistency, the model was re-trained with the previously selected optimal hyperparameter value of `n_estimators` = 500, while all other parameters remained constant from the tuning phase.

To ensure successful learning, the re-trained model’s performance was tested on the entire training data set. The final model had a R^2 of 0.9356, an MSE of 22.56, and an MAE of 3.49. As expected, these metrics show a strong fit to the combined training data, reflecting the increased quantity of information available to the model.

This re-trained model is used as the final predictive model, and it is then tested on the remaining test set to get a fair estimate of its generalisation performance.

5 Evaluation

During evaluation, the final Random Forest model (`estimators` = 500) was first retrained on the combined training and validation sets, and then applied exactly once to the test set. This follows standard practice to maximize the data available for learning, but also getting an unbiased estimate of generalization performance.

The retrained model achieved very strong performance on the combined training data ($R^2 = 0.936$, $MSE = 22.56$, $MAE = 3.49$), which is expected, because the model has already seen these observations during training. The more informative picture comes from the independent test evaluation.

On the held-out test set, the final model obtained a mean absolute error (MAE) of 10.78, a root mean squared error (RMSE) of 14.60, a mean squared error (MSE) of 213.29 and a coefficient of determination R^2 of 0.456. Since graduation rate is measured in percentage points, an MAE of around 10.8 means that, on average, the model’s prediction deviates by roughly eleven percentage points from the true graduation rate. An R^2 value of approximately 0.46 indicates that the model explains about 46% of the variance in graduation rates across institutions. This represents a moderate but meaningful level of predictive power in a noisy, real-world dataset with substantial missing and diverse value among institutions.

To provide a compact overview and to satisfy the requirement of comparing against a trivial baseline, Table 8 summarizes the test performance of the final Random Forest model and of a naive predictor that always outputs the mean graduation rate observed in the training data.

Table 8: Test-Set Performance: Final Model vs. Trivial Baseline

Model	MAE	RMSE	MSE	R^2
Random Forest (final)	10.78	14.60	213.29	0.456
Mean baseline	16.25	19.80	391.96	≈ 0.00

The table shows that if compared to the trivial mean predictor, the Random Forest regressor clearly improves performance: the MAE is reduced by about 5.47 percentage points (from 16.25 to 10.78), and the MSE is almost halved (from 391.96 to 213.29). At the level of explained variance, the baseline accounts for essentially none of the variation in graduation rates, whereas the final model achieves $R^2 = 0.456$. This shows that the model captures substantial structure in the data beyond a constant prediction and achieves the requirement to compare against a trivial baseline.

To address the requirement of identifying state-of-the-art performance, we searched for published benchmark studies using the chosen dataset in particular for regression tasks predicting graduation rate. We did not find peer-reviewed work reporting comparable numeric results for this specific prediction task. The dataset appears to be used mainly for teaching, exploratory data analysis and demonstration purposes rather than as a standard benchmark for graduation-rate regression. In the absence of external numeric baselines, we treat our tuned Random Forest model as an internal benchmark and focus on the comparison to the trivial mean predictor and to the success criteria defined in the Business Understanding phase.

From a business and data-mining perspective, the success criteria required that the model should explain a substantial fraction of the variance in graduation rates and achieve a reasonably low prediction error, while remaining interpretable enough to support decision-making. With an R^2 of 0.456 on the test set and an MAE of 10.78 percentage points, the model partially fills these criteria. It does significantly better than a naive baseline and highlights

relevant predictors of graduation rate, but it does not achieve perfect accuracy, which is realistic given the complexity of the underlying process and the limitations of the available data. Overall, the results are consistent with the expectations for this type of institutional-level dataset and are sufficient to support exploratory, decision-support use rather than high-stakes automated decision-making.

To evaluate potential bias, the `Public/private_indicator` was selected as a “protected” attribute that distinguishes between public and private institutions. Using the original cleaned dataset, we computed error metrics separately for both groups on the test set. For private institutions ($n = 59$), the model achieved $MAE = 9.29$, $RMSE = 11.48$ and $R^2 = 0.511$. For public institutions ($n = 122$), the corresponding metrics were $MAE = 11.51$, $RMSE = 15.90$ and $R^2 = 0.341$. Thus, the model is more accurate and explains more variance for private colleges than for public ones. The difference in MAE of about 2.2 percentage points and the gap in R^2 suggests a moderate performance difference between these groups.

This performance difference can have several explanations. On the one hand, private institutions may have more homogeneous characteristics (e.g., tuition level, selectivity, spending per student), which makes their graduation rates easier to model. Public institutions, by contrast, cover a much wider range of sizes, missions and funding structures, which may lead to greater variability that the current model cannot fully capture. On the other hand, the observed difference also indicates a potential bias in the model’s error distribution: predictions for public institutions tend to be less accurate than for private ones. A more detailed fairness analysis, including explicit fairness metrics and consultation with domain experts, would be necessary before using the model in any context where such differences could lead to disadvantageous treatment of a particular institutional subgroup.

In summary, the evaluation phase shows that the final Random Forest model achieves a clear and substantial improvement over a trivial baseline and provides a reasonable predictive approximation of graduation rates, while still exhibiting moderate generalization errors and unequal performance across institutional types. These findings underline that the model is suitable as an analytical and decision-support tool, but not as an automated decision-making system. They also motivate potential future work on model refinement, such as incorporating additional external data sources, experimenting with alternative algorithms, or addressing fairness constraints to reduce performance differences between public and private institutions.

6 Deployment

6.1 Comparison and Recommendations

The evaluation results show that the final Random Forest model partially meets the business success criteria defined in the business understanding phase. On the one hand, the model achieves a clear improvement over a trivial mean baseline on the test set (MAE reduced from 16.25 to 10.78 percentage points, MSE almost halved, and R^2 increasing from approximately 0 to 0.456). This indicates that the model captures substantial structure in the data and explains a meaningful part of the variance in graduation rates.

On the other hand, the remaining prediction error of around eleven percentage points MAE, and an R^2 clearly below 1.0, show

that the model is far from perfectly accurate. Given the complexity and noisiness of institutional data, this is not unexpected, but it implies that the model should not be used as the sole basis for high-stakes decisions (such as funding allocation or detailed performance evaluation of individual institutions). Instead, it is better suited as an analytical support tool to identify patterns, compare institutions at an aggregated level, and explore potential scenarios, consistent with the business objectives of benchmarking, understanding factors of graduation rate and informing strategy rather than fully automating decisions.

We recommend a hybrid decision-support setup. The model could be integrated into a dashboard where institutional analysts can inspect predicted graduation rates, feature importance, and group performance. If a more detailed deployment is desired, it would be reasonable to restrict automated usage to parts of the data space where model performance is known to be stable (for example, institution types and size ranges with sufficient training data), while flagging predictions in more uncertain regions for manual review. Further analyses that would strengthen deployment include: (i) experimenting with alternative algorithms and feature engineering (e.g. derived selectivity and total cost measures), (ii) integrating external socio-economic data to improve explanatory power, and (iii) performing a more systematic fairness and robustness analysis across additional subgroups.

6.2 Ethical Aspects

The deployment of a predictive model for graduation rates involves several ethical aspects that build directly on the AI risk considerations formulated in the business understanding phase and the sensitivity analysis in the data understanding phase. First, many variables in the data set (such as public vs. private status, spending per student, or selectivity measures) reflect structural inequalities between institutions. If model outputs were used without a prior critical analysis for rankings, funding decisions, or public communication, there is a risk of reinforcing existing advantages of institutions that are well resourced and penalizing those with less preferable conditions. The evaluation phase also showed moderate performance differences between public and private institutions, with better accuracy for private colleges, which may further indicate unequal treatment if not explicitly addressed.

Second, there is a danger of treating predicted graduation rates as objective quality indicators. The model is trained on historical data and correlations. It does not capture all relevant factors (such as regional labour markets, social and economic background of students, or institutional missions) and does not provide causal explanations. Using such predictions to make significant decisions (closing programmes, changing admissions policy, or reallocating funding) without careful expert review would be ethically problematic. The earlier analysis in 1.f) and 2.e) already highlighted the risk of over-interpreting correlations as causal and of embedding the original US News ranking biases into downstream decision-making.

In light of these risks, deployment should explicitly frame the model as a decision-support tool rather than an automatic decision system. Predicted values and feature importance measures should be communicated together with uncertainty and limitations, and

stakeholders should be cautioned against interpreting them as definitive judgements of institutional quality. Any use of the model in policy or funding contexts should be accompanied by transparency about the data sources, modeling assumptions, and known fairness limitations, and should include mechanisms for contesting or overriding model-based suggestions, especially in cases where disadvantaged institution types might be negatively affected.

6.3 Monitoring Plan

To ensure the continued reliability and ethical use of the deployed graduation rate prediction model we need to ensure monitoring of the predictive performance, data drift and fairness across key subgroups. To do this, first the model performance on newly arriving data should be tracked using the same metrics as in the evaluation phase (MAE, RME and R^2). We can evaluate how the model is performing by comparing the future metrics with the baseline established during the current experiment. If we notice a significant degradation on these metrics, that would indicate that the relationship between predictors and graduation rates has changed which means that there is a concept drift or that quality issues have emerged in the data.

Second, we need to monitor the input data distribution needs. Which means tracking basic, descriptive statistics and missing-value patterns for key features such as SAT/ACT scores, cost variables, and enrollment numbers. Furthermore, we need to monitor changes in the percentage of public vs. private institutions or different size classes. By investigating these aspects for large distributions shifts or sudden changes in the frequency of missing values, we make sure that the model remains applicable or whether retraining and possible re-design of preprocessing steps are required.

Third, fairness and bias aspects should be monitored over time. Error metrics consisting of MAE, RMSE, R^2 should be calculated separately for each new evaluation period. If over time from investigating these values we exam that the difference in errors between these groups becomes too large, such as MAE differing by more than a few percentage points, this means that the features used for this model and how the predictions are applied in decision making need a closer review.

Additionally, it is important to record cases where model's predictions are overridden by human experts and whether these overrides follow clear patterns. These pattern can help indicate weaknesses in the model or ethical concerns.

To conclude these monitoring activities provide clear triggers for intervention, such as retraining the model, adjusting the feature set, limiting the model's use to certain cases, or, in extreme situations, temporarily stopping the use of the model until the issues are better understood.

6.4 Reproducibility

This experiment was developed with reproducibility in mind, and this can be seen in many factors, such as the use of a publicly available dataset (colleges_usnew), a clearly defined conda environment with an requirements file and fixed number of seeds which was used for data splitting and model training. The whole workflow is implemented in a notebook with clearly separated steps for data loading, data preprocessing, modeling and evaluation. Furthermore,

the provenance knowledge graph records which activities were carried out, when they happened, and which data, models and results were involved.

With all that being said this experiment has also some limitations regarding reproducibility. Reproducibility depends on the continued availability and stable behaviour of external services such as OpenML and the provenance database. Another factor that can impact reproducibility is changes in software versions, for example changes in Python or scikit-learn may lead to slightly different numerical results. Furthermore some parts of the Data Understanding and Data Preparation phases required manual inspection, such as identifying outliers and deciding not to remove them and even though these decisions are documented they introduce a degree of subjectivity that cannot be fully automated.

All in all, we can confidently say that the experiment is reproducible in terms of its main steps and conclusions. If another analyst follows the documented process and uses the same dataset with similar software versions should obtain comparable results and insights. With that being said, exact reproduction of every numeric result cannot be fully guaranteed based on the report and provenance information alone, because details such as exact library versions, hardware, and the state of external services are not fully controlled.

7 Conclusion

Working through the full CRISP-DM cycle on a real-world dataset highlighted how strongly the different phases of a data analytics project depend on each other. Early choices in the Business Understanding and Data Understanding phases, such as defining graduation rate as the primary target and success criteria, directly impacted preprocessing, modeling, and evaluation. Having clear criteria early helped us interpret the final results, where the model achieved moderate predictive performance with an R^2 of 46 and an MAE of 10.8. These results show that the model is appropriate for exploratory decision support rather than automated decision-making.

From a modeling perspective, the experiment demonstrated both the strengths and limits of Random Forests on noisy, heterogeneous institutional data. Although the final model clearly outperformed a trivial baseline and captured meaningful structure in the data, a substantial portion of the variance in graduation rates remained unexplained, underlining the complexity of the underlying process.

From a technical and organizational perspective, the exercise highlighted the importance of reproducibility and structured documentation. Using fixed data splits, a controlled environment, and consistently logging activities and entities into the provenance knowledge graph all added overhead but also increased transparency and made the workflow easier to understand and reproduce. Finally, evaluating performance across public and private institutions showed that acceptable overall accuracy can still hide subgroup differences, reinforcing the need to consider fairness and ethical aspects alongside predictive performance.

Overall, the exercise reinforced that successful business intelligence projects are not only about achieving good predictive accuracy but also about building transparent, well-documented, and responsible analytical processes that can be understood and critiqued by stakeholders beyond the data science team.